

Baker College Teacher Prep Lesson Plan Format

Subject Area & Grade Level: Geometry Grade 3

Lesson Duration: 25 minutes

<u>Lesson Goal:</u> What do we want students to learn?	<u>Assessment:</u> How will we know if they have learned it? (see guide)	<u>Intervention:</u> What will we do if they don't learn it? (see guide)	<u>Enrichment:</u> What will we do if they already know it? (see guide)
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Common Core State Standard:

CCSS: 3.G.1, 3.G.2 – Geometry:

Reason with shapes and their attributes. As the Michigan Department of Education (n.d.) outlines on page 28, students should understand that shapes in different categories may share attributes and that shared attributes can define a larger category. Students will identify and distinguish properties of rectangles, squares, and triangles by folding paper into each shape without using the edges of the paper, and convince a skeptical partner using geometric reasoning.

Learning Objective:

(In words that your students would understand)

I can fold paper into a rectangle, square, and two different triangles without using the edges of the paper. I can explain *why* each shape is what it is using math words like right angles and equal sides, and I can ask good questions when my partner is trying to convince me.

Formative:

Teacher observes partner conversations during the Explore phase, listening for use of geometric vocabulary and quality of reasoning. An exit ticket at the end of the Discuss phase asks students to write one thing they used to convince their partner and one question they asked as a skeptic.

Intervention:

☑ Differentiated Instruction – Instruction is adjusted based on student readiness with the geometric vocabulary and reasoning demands of the task.

☑ Target specific skills – Students who cannot articulate shape properties are targeted for direct support on angle and side-length vocabulary.

☑ Leveled materials – A visual word bank is provided below grade level; on-level students use the standard worksheet; above-level students are pressed to write full written arguments.

☑ Graphic organizers – Students use a reasoning organizer to structure their convincing argument before speaking to their partner.

☑ Manipulatives – Angle checkers are available for students who need a physical tool to verify right angles.

☑ Immediate feedback – Teacher circulates during Explore and

☑ Independent projects – Students who demonstrate mastery early are challenged to fold a fourth shape of their choice and write a complete written argument convincing a skeptic.

☑ Peer tutoring – Mastery students support struggling partners by modeling strong skeptic questioning rather than giving answers.

☑ Group leader – Students serve as discussion leaders during the whole-class Discuss phase, posing skeptical questions to other pairs and facilitating reasoning.

☑ Use vocabulary to write sentences – Students write a full geometric argument using shape-property vocabulary to justify each of their folds.

		provides in-the-moment corrective feedback. ☑ Flexible grouping – The teacher pulls a small group during Explore for guided questioning. Rtl: ☑ Small group instruction – Teacher-led small group during Explore for students not yet accessing geometric reasoning. ☑ Re-teach in a different way – If students cannot reason about angles verbally, the teacher demonstrates using the angle checker manipulative as an alternative entry point. ☑ Modify: backtrack, build background knowledge – Students who lack prerequisite knowledge of shape names are briefly re-grounded in defining attributes before rejoining partner work.	
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Materials:

- ☐ One piece of 8.5" × 11" paper per student
- ☐ Paper Folding worksheet, one per student
- ☐ Visual vocabulary word bank covering right angle, parallel, equal sides, and vertices — teacher-created
- ☐ Angle checker manipulatives, optional, for intervention
- ☐ Whiteboard and projector for Discuss phase

Select your learning strategy:

? ☒ **Cooperative Learning**

? ☒ **Discovery/Inquiry-Based Learning**

- Direct Teach
- Demonstration
- Cooperative Learning
- Discovery/Inquiry-Based Learning
- Project-Based Learning
- Reading/Writing/Math Workshop
- Other

Activities Planned: ☒ Active (Students are active participants in learning)

☐ Passive (Teacher led lecture/demonstration)

☐ Both

Lesson Delivery Steps:

As Boaler et al. (2018) explain on page 24, one of the most important topics in mathematics is reasoning, and this activity gives students an opportunity to learn to reason by having to convince others who are being skeptical.

Snapshot

Students work in pairs to fold paper into a rectangle, square, and two different triangles — none using the edges of the paper — while alternating between the roles of convincer and skeptic. The mathematical work is in the reasoning, not just the folding.

Connection to CCSS

The Michigan Department of Education (n.d.) identifies on page 28 that third-grade geometry standards require students to reason with shapes and their attributes, including recognizing rhombuses, rectangles, and squares as examples of quadrilaterals and drawing examples of quadrilaterals that do not belong to any of those subcategories.

Agenda

Activity Time Description

Launch 5 min Introduce convincer and skeptic roles; model with a volunteer

Explore 10 min Partner paper folding across four challenges

Discuss 10 min Whole-class debrief on reasoning and skepticism

To the Teacher

Boaler et al. (2018) note on page 24 that before beginning the activity, the teacher should explain to students that their role is to be convincing, and that the highest level of being convincing is to convince a skeptic. Students need to understand that saying it looks like it is not an acceptable mathematical argument. Encourage students to use shape properties such as angles, side lengths, and parallelism as the basis of their reasoning. The teacher's role during Explore is to circulate and press students by asking how they know the angles are all right angles.

Activity

Launch (5 minutes)

Tell students that their role for the day is to be convincing and to be a skeptic. Boaler et al. (2018) describe on page 25 that students should work in pairs and take turns being the one convincing and the one being skeptical. Ask students to fold a piece of paper into a rectangle that is not a square. Choose a student volunteer and model what it looks and sounds like to be a skeptic, asking questions such as how they know the angles are all right angles rather than accepting that it looks correct.

Explore (10 minutes)

Show students the four challenges and explain that in each round they alternate folding and reasoning and being the skeptic. After students convince themselves they have solved each problem, they switch roles and fold the next challenge. Boaler et al. (2018) explain on page 25 that partners must reason and be skeptical for each challenge. Give each student square paper or ask them to make a square first. The four challenges are:

1. Fold your paper into a rectangle that does not include any edges of the paper. Convince a skeptic that it is a rectangle. Reflection / Switch roles.
2. Fold your paper into a square that does not include any edges of the paper. Convince a skeptic that it is a square. Reflection / Switch roles.
3. Fold your paper into a triangle that does not include any edges of the paper. Convince a skeptic that it is a triangle. Reflection / Switch roles.
4. Fold your paper into a different triangle that does not include any edges of the paper. Convince a skeptic that it is a triangle. Reflection.

Discuss (10 minutes)

Facilitate a whole-class discussion. Boaler et al. (2018) suggest on page 25 that the class discussion should address which task was most challenging and why, what was hard about reasoning and being convincing, and what was hard about being a skeptic. Use this time to reinforce geometric vocabulary and highlight exemplary reasoning heard during Explore. Record student language on the board to build a class anchor chart of convincing mathematical arguments.

Look-Fors (Teacher Observation Focus)

- Are students using geometric properties such as angles and side lengths rather than visual appearance to convince?
- Are skeptics asking substantive questions rather than simply accepting or rejecting?
- Are students able to distinguish a square from a rectangle and articulate why?
- Which students may need vocabulary support or small-group intervention?

Reflect

Exit ticket: Students write one sentence describing what they used to convince their partner and one sentence describing a question they asked as a skeptic. Teacher reviews responses to identify students needing re-teaching of geometric properties before the next lesson.

Check the Core Teaching Practices addressed in your lesson: (from MDE CTPs)

- Leading a group discussion (CTP #1)
- Explaining and modeling content, practices, and strategies (CTP #2)
- Eliciting and interpreting individual students' thinking (CTP #3)
- Building respectful relationships with students (CTP #10)
- Check for Understanding (CTP #15)

☐ ☒ Leading a group discussion (CTP #1) — whole-class Discuss phase

☐ ☒ Explaining and modeling content, practices, and strategies (CTP #2) — Launch modeling of skeptic role

- ☐ ☒ Eliciting and interpreting individual students' thinking (CTP #3) — circulating during Explore; exit ticket
- ☐ ☒ Building respectful relationships with students (CTP #10) — partner roles built on mutual respect; no idea is dismissed without a mathematical reason
- ☐ ☒ Check for Understanding (CTP #15) — exit ticket; teacher observation during Explore

List real-world connections including attention to English language learners and culturally and historically responsive practices (diversity, inclusion, equity, and social justice):

This activity connects geometry to the real world through architecture, origami, and engineering, where shapes and their properties must be precisely reasoned and communicated. The Michigan Department of Education (n.d.) emphasizes on page 28 that students should be able to recognize and draw examples of shapes and explain shared attributes, which mirrors the kind of reasoning professionals use in design fields. The cooperative structure ensures all voices are heard — students take turns as both convincer and skeptic, so no single student dominates. The activity is accessible regardless of language background because the physical paper folding provides a concrete, visual referent. The teacher-created vocabulary word bank supports English Language Learners by pairing geometric terms with diagrams. The emphasis on reasoning over right answers creates an equitable mathematical space where students with varied prior knowledge can contribute meaningfully.

List technology tools:

- ☐ Document camera to display student paper folds during Discuss phase
- ☐ Projector and whiteboard to record class anchor chart of convincing arguments

List collaboration opportunities (whole group, small group, partnerships, building resource personnel i.e., school social worker, special educators, parents, etc.):

- ☐ Partnerships: Core structure of the activity — every student works with a partner, alternating convincer and skeptic roles. Boaler et al. (2018) describe on page 24 that students should work in pairs and take turns being the convincer and the skeptic.
- ☐ Whole group: Whole-class Discuss phase
- ☐ Small group: Teacher-led small group during Explore for students needing intervention

- ☐ Special educator / paraprofessional: Can co-circulate during Explore, supporting students with IEPs on geometric vocabulary and verbal expression
- ☐ Parents: Families can extend the activity at home by repeating the folding challenges with household paper

References

Boaler, J., Munson, J., & Williams, C. (2018). *Mindset mathematics: Visualizing and investigating big ideas, grade 3*. Jossey-Bass.

Michigan Department of Education. (n.d.). *Michigan K-12 standards: Mathematics*.

https://www.michigan.gov/-/media/Project/Websites/mde/Literacy/Content-Standards/Math_Standards.pdf?rev=1e793e2b1e314e4fa1abc754251b5dc9

Lesson Plan Guide

Assessment: Used to gather information about a student's progress towards mastery of the learning objective, help the teacher identify what instruction is working well and what needs refinement, and informs the students about their learning.

Options to consider

- ☐ Diagnostic/Pre-Assessment – Used to check prior knowledge **before** a lesson
- ☐ Self-Assessment (Writing Prompts, Running Records, Performance Task, Other)
- ☒ Formative – Used **during** a lesson to check progress, identify any misconception, and give feedback to students (Learning/Response Log, Admin/Exit Ticket, Think/Pair/Share, One Minute Paper, Other)
- ☐ Summative – Used at the **end** of a lesson to check student mastery of the objective (End of Unit Test, Final Exams or Mid-term Exams, State Tests, Culminating Project, Portfolio, Other)

Intervention: How will we respond when they don't learn?

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| ☒ Differentiated Instruction | ☐ Referral to Student Support Team |
| ☒ Target specific skills | ☒ Graphic organizers |
| ☐ Data item analysis | ☒ Manipulatives |
| ☒ Leveled materials (below, on level, above) | ☐ Choice boards |
| ☐ Bloom's Taxonomy | ☒ Immediate feedback |
| ☐ Grade recovery (re-do/correct) | ☒ Flexible grouping |
| ☐ Parent contact | ☐ Extended responses (math/reading) |
| | ☐ Journal/Reading logs |

Responses to Intervention (RtI)

- ☒ Small group instruction
- ☐ Tiered group instruction (Tier I, II, III)
- ☐ 1-1
- ☐ Centers (leveled)
- ☒ Re-teach in a different way
- ☒ Modify: backtrack, build background knowledge
- ☐ Tutoring: after or before school, lunch
- ☐ Referral to Student Support Team

Enrichment: How will we respond if they already know it?

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| ☐ Choice boards | ☒ Group leader |
| ☒ Use vocabulary to write sentences | |
| ☐ Accelerated reader | |
| ☐ Centers-High level | |
| ☐ Reading buddies | |
| ☒ Peer tutoring | |
| ☐ Enriched-Leveled Reader-Novels | |
| ☐ Picture/writing journals | |
| ☒ Independent projects | |
| ☐ Separate curriculum | |
| ☐ Games | |